



## Playing by the (policy) rules: Why ECB tightening is warranted

The ECB has embarked on a path of measured policy tightening, with President Lagarde [describing](#) June's 25bp hike as "robust" but not "forceful." To assess the appropriate trajectory of policy and the potential quantum of further hikes, we have applied a suite of simple policy rules to the ECB's own staff projections. This report shows that the ECB's own outlook for inflation and growth warrants some further policy tightening ahead, even after the recent US-Iran MOU announcement.

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### Conclusions and takeaways:

- **Further policy tightening is warranted:** The ECB's June rate hike was robust, supported even by the milder scenario. Using the ex-energy HICP forecasts, the median policy prescription under the baseline scenario is consistent with rates rising to roughly 2.75% by end-2026 (consistent with two more hikes). Or under the milder scenario, the median prescription suggests rates topping out at a little less than 2.50% at the end of this year (more or less consistent with one more hike).
- **June's revisions to the baseline projection warrant some further policy tightening:** The revision to inflation and activity between the March baseline projection and the June projection boost the prescribed path of policy rates by 25bp to 50bp on average across the various rules. Between March and June, market pricing moved to price in an additional 25bp policy hike this year – broadly consistent with the policy rules.
- **Rules are only a guide not a script:** While policy rules provide a guide for "good" monetary policy, they do not capture all the complexities of monetary policy. During the pandemic and energy shock from Russia's invasion of Ukraine, the rules prescribed highly aggressive policy tightening. During the 2024 rate cutting cycle, actual rates broadly followed the prescribed paths for rates early on, but perhaps pushed actual rates towards the lower end of the prescribed range by the end of the cutting cycle.

**What this means for the ECB policy outlook:** The tone at the ECB press conference on 11 June was consistent with the risk of two more hikes, leading to a terminal policy rate of 2.75%. We maintain our baseline call for one more hike to 2.50%. There are already some signs of the inflation shock moderating (e.g., [dbDIG inflation expectations](#)) and oil prices are lower than what we were expecting after a US-Iran deal. Current energy futures suggest the path ahead lies somewhere between the ECB's baseline and milder scenarios. If that is the case, the policy rules



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are consistent with another one or two hikes.



## Methodology

This report examines what interest rates should do according to a range of simple policy rules. The suite of policy rules is taken from a set regularly monitored by the Federal Reserve (see [here](#) and [here](#)) and includes a range of different rules based on various common measures of inflation (headline HICP, core HICP) and activity (unemployment, the output gap). The table in [Figure 1](#) details the policy rules used.

Figure 1: The suite of policy rules considered

| Rule                                 | Equation                                                                                                                        |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Taylor (1993) rule                   | $i_t = r_t^{LR} + \pi_t + 0.5(\pi_t - \pi^{LR}) + (u_t^{LR} - u_t)$                                                             |
| Balanced-approach rule               | $i_t = r_t^{LR} + \pi_t + 0.5(\pi_t - \pi^{LR}) + 2(u_t^{LR} - u_t)$                                                            |
| Taylor (1993) rule with output gap   | $i_t = r_t^{LR} + \pi_t + 0.5(\pi_t - \pi^{LR}) + \text{output gap}_t$                                                          |
| Forward-looking rule                 | $i_t = r_t^{LR} + \pi_{t+3}^F + 0.5(\pi_{t+3}^F - \pi^{LR}) + 0.5(\text{output gap}_t)$                                         |
| Core inflation in Taylor (1999) rule | $i_t = r_t^{LR} + \pi_t^{\text{core}} + 0.5(\pi_t^{\text{core}} - \pi^{LR}) + \text{output gap}_t$                              |
| First-difference                     | $i_t = i_{t-1} + 0.5(\pi_t - \pi^{LR}) + (u_t^{LR} - u_t) - (u_{t-4}^{LR} - u_{t-4})$                                           |
| First-difference (ver. 2)            | $i_t = i_{t-1} + 1.74(\pi_t - \pi^{LR}) - 1.19(u_{t-1} - u_{t-2})$                                                              |
| Inertial rule                        | $i_t = 0.8i_{t-1} + (1 - 0.8)[r_t^{LR} + \pi_t^{\text{core}} + 0.5(\pi_t^{\text{core}} - \pi^{LR}) + \text{output gap}_t]$      |
| Low weight on output gap rule        | $i_t = 0.91i_{t-1} + (1 - 0.91)[r_t^{LR} + \pi_t^{\text{core}} + 1.58(\pi_{t+1}^F - \pi^{LR}) + 0.14(\text{output gap}_{t+1})]$ |

Source : Deutsche Bank Research, Federal Reserve. Notes: Colouring group rules based on the "type" of rule

### The rules:

- **Taylor rule:** The rule is inspired by the original 1993-version of Taylor's rule. Nominal interest rates should be raised by 1.5 times the inflation gap (more than one-for-one in order to satisfy the Taylor principle). Rates should also be raised when the unemployment rate is below its long-run value.
- **Balanced-approach rule:** This rule modifies the previous rule to place more importance on the unemployment gap. Although the ECB does not have a dual mandate (like the Fed), because of the lags in inflationary pressure, targeting the current unemployment gap has similarities to also targeting future inflation (which unemployment affects through the Phillips curve).
- **Taylor rule with output gap:** This rule replaces the unemployment gap with the output gap, an alternative measure of the slack in the economy.
- **Forward-looking rule:** This rule targets future inflation rather than current inflation, which can help the central bank look through near-term variability they cannot affect due to the lags of monetary policy.
- **Core inflation in Taylor rule:** An alternative approach to looking through near-term variability is to replace headline inflation with core inflation in the policy rule.
- **First-difference rules:** These rules say the change in the policy rate should be related to inflation gaps and changes in the unemployment rate. If either of these two terms are non-zero, it suggest the economy is not in equilibrium, and policy should be adjusted to compensate. Version 2 does not rely on the long-run unemployment rate (which is unobservable), and could thus be more robust in real-time policy setting.
- **Inertial rule:** This rule adds policy smoothing to the rule in an attempt to



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better replicate the gradual nature of policy adjustments .

- **Low weight on output gap rule:** This inertial-type rule lowers the weight on the output gap and increases it on inflation to focus more on inflation.

For forecasts of the variables in the rules we use the ECB's latest set of staff projections (released at the June meeting).<sup>1</sup>

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## Policy rule prescription for future interest rates

We begin the analysis by considering the central question: what do the policy rules suggest the ECB should do with interest rates over the near term?

Feeding the ECB's own June staff projections into the suite of policy rules, we have produced the path of interest rates prescribed by each rule over 2026-27 ([Figure 2](#), LHS).

The suggested paths of interest rates broadly fall into three main groupings:

- **Taylor-type rules predicated on headline HICP inflation** suggest monetary policy should be tightened aggressively and quickly in response to the inflationary pressures from the energy price shock. The rules suggest rates should be hiked to around 4% by the end of this year, a significant increase from current levels.
- **First-difference rules** prescribe a steady tightening of policy. While less aggressive than the Taylor-type rules the fact that inflation is well above target in 2026, and further on unemployment is steadily falling (driven by the economic recovery) the rule interpret the data as signals we are still below equilibrium throughout this period.
- **Policy rules that incorporate smoothing of policy rates or based on core inflation** suggest policy paths that are more in line with what markets and economists (the SMA) are predicting. The smoothing parameter helps the rule look through the near-term inflationary spike that is assumed in the forecasts to be temporary and take a more medium-term view.

**The ECB can probably ignore the signal from policy rules based on headline inflation in an energy supply shock. Policy can do nothing about the direct energy effects and will instead focus on the spillovers or second-round effects from energy prices. For this reason, we also run the rules on HICP excluding energy as the inflation measure within the rules.** These results are presented on the right-hand side of [Figure 2](#).

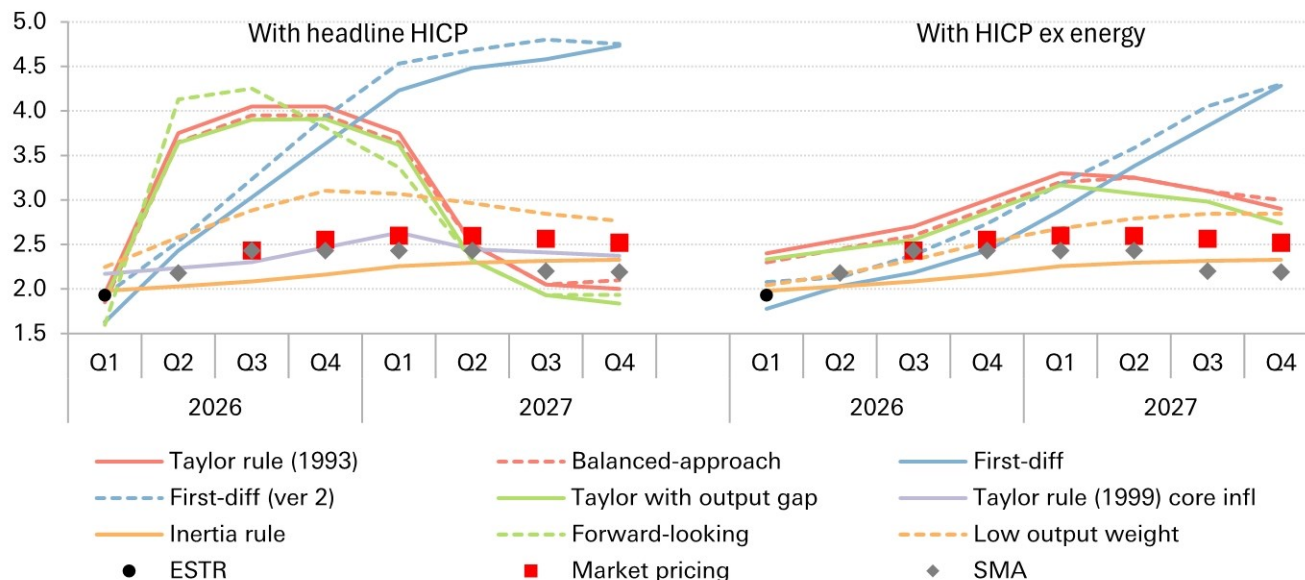
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<sup>1</sup> For some rules we require estimates of unobservable components such as potential output, neutral rate, and long-run unemployment rate. Where possible, the average estimates of the long-run values of variables from the ECB's Survey of Monetary Analysts (SMA) are used as the estimates of the long-run or neutral rates in the rules. When not possible, statistical filters are used.





Figure 2: Path of policy rates (ESTR) prescribed by a suite of monetary policy rules produced using the ECB's June staff baseline projections



Source : Deutsche Bank Research, ECB, Haver Analytics, Bloomberg Finance LP  
Notes: SMA = Survey of Monetary Analysts. Colour of lines group rules based on the "type" of rule

Excluding energy prices from the inflation measure produces policy prescriptions that lie much closer to current market pricing. The rules suggest that further tightening of policy is warranted under the ECB's baseline projects which incorporate a broadening of inflation throughout the economy. **The median rule implies rates at roughly 2.75% by end-2026 (consistent with two more hikes).**

**Caveat for interpreting the policy prescriptions:** The path of inflation and GDP used in the policy rules are drawn from the staff's baseline projections which are conditioned on market pricing (three interest rate hikes). There is no endogenous feedback from the path of prescribed policy from each policy rule back to inflation and output. In reality, if the ECB did follow the Taylor-type rules prescription at the last meeting and hike to around 3.75% rather than 2.25%, the future path of inflation and GDP growth would be different from what is assumed in the baseline projections. This in turn would affect the values of the prescribed policy from the rule further out in the projection horizon.

## How to respond to the alternative scenarios

The ECB staff projections introduced a new milder scenario to the downside of their baseline projection. With the recent announcement of the memorandum of understanding (MOU) between the US and Iran, the spot prices of oil and gas have fallen significantly, and with them the near-term energy futures as well. **Over the rest of 2026, energy futures now lie closest to the ECB's milder scenario. However, further out in the forecast horizon, energy futures remain closer to the baseline projection (Figure 3).**

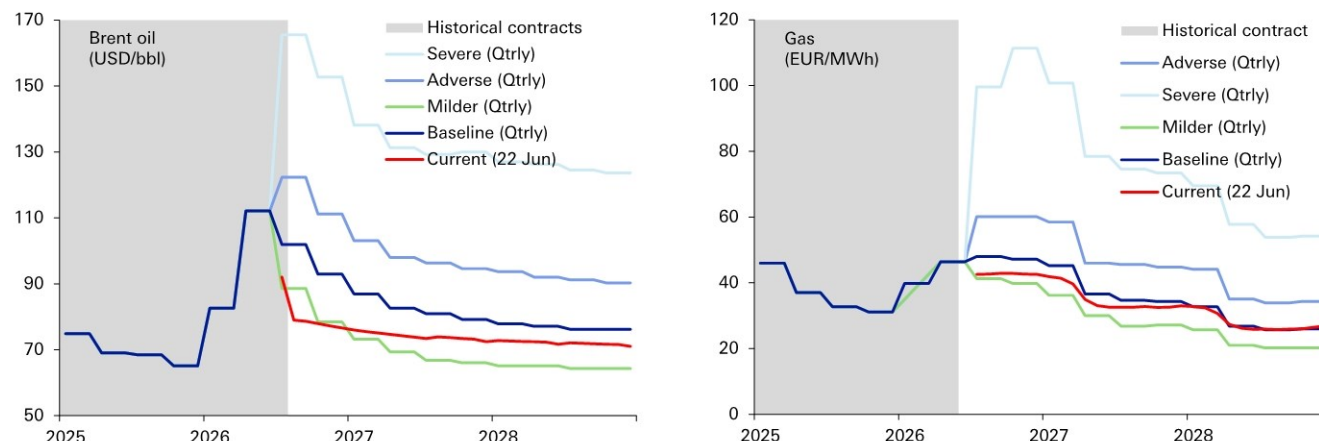
During the [press conference](#), Lagarde said that "the decision to raise rates is robust across a range of scenarios mapping out how the shock might evolve and affect the medium-term outlook for the euro area".



We have re-run the policy rules on all three alternative scenarios (Milder, Adverse, and Severe) to see how the policy prescription of each rule would differ from the baseline.

One important caveat is that the alternative scenario projections published by the ECB do not include their forecast for unemployment. Therefore, we can only analyse how some of the rules differ across scenarios – those that rely on the output gap rather than unemployment.

Figure 3: Oil and gas futures contracts vs the ECB's June alternative scenarios



Source : Deutsche Bank Research, ECB, Bloomberg Finance LP

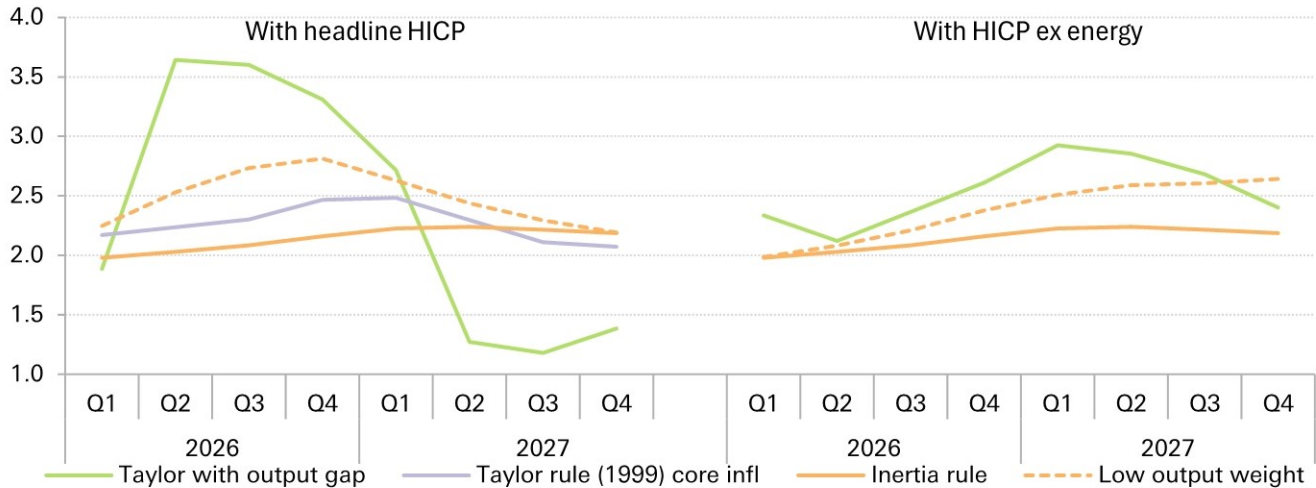
Figure 4 to Figure 6 shows the policy prescriptions of the various rules under the alternative scenarios the ECB published.

- **Milder scenario:** The median prescribed interest rate from the policy rules (excluding energy from HICP) is consistent with policy rates around 2.50% from the end of 2026 onwards (consistent with one more hike this year).
- **Adverse and severe scenarios:** Both rules have similar prescriptions to the baseline and milder scenario over 2026. This is because the alternative scenarios only begin to differ from Q3 2026, and it takes time for the higher energy prices under these scenarios to filter through to HICP excluding energy prices. But over 2027, both rules prescribe further steady tightening of policy throughout the year.

The ECB's claim that the June hike was robust is supported by the analysis of this framework. **Even in the milder scenario, the rules suggest the 25bp hike in June was justified under the ECB's staff projections, and one further hike (to 2.50%) could also be appropriate.** The fact that interest rates in the adverse and severe scenarios begin to deviate significantly only in 2027 suggests the ECB should be happy to leave some tightening risk premia in the market after it hikes to 2.50% in September – if there is any risk of the adverse or severe scenarios still materialising.



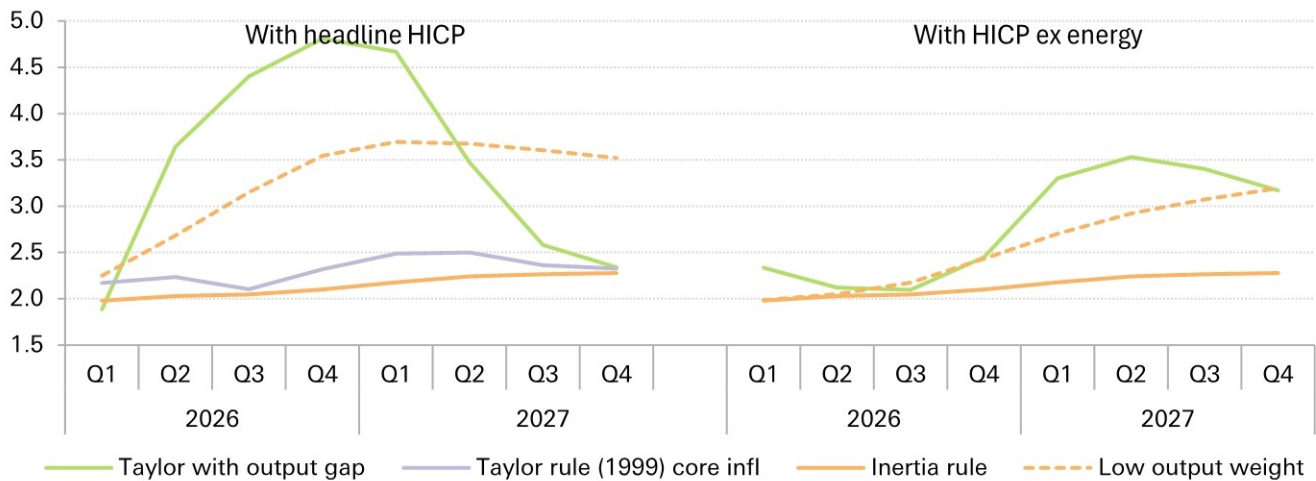
Figure 4: Mild scenario – policy rule prescriptions



Source : Deutsche Bank Research, ECB, Haver Analytics, Bloomberg Finance LP.

Notes: Only a subset of policy rules can be analysed as the ECB does not publish unemployment data for their alternative scenarios. Colour of lines group rules based on the "type" of rule.

Figure 5: Adverse scenario – policy rule prescriptions



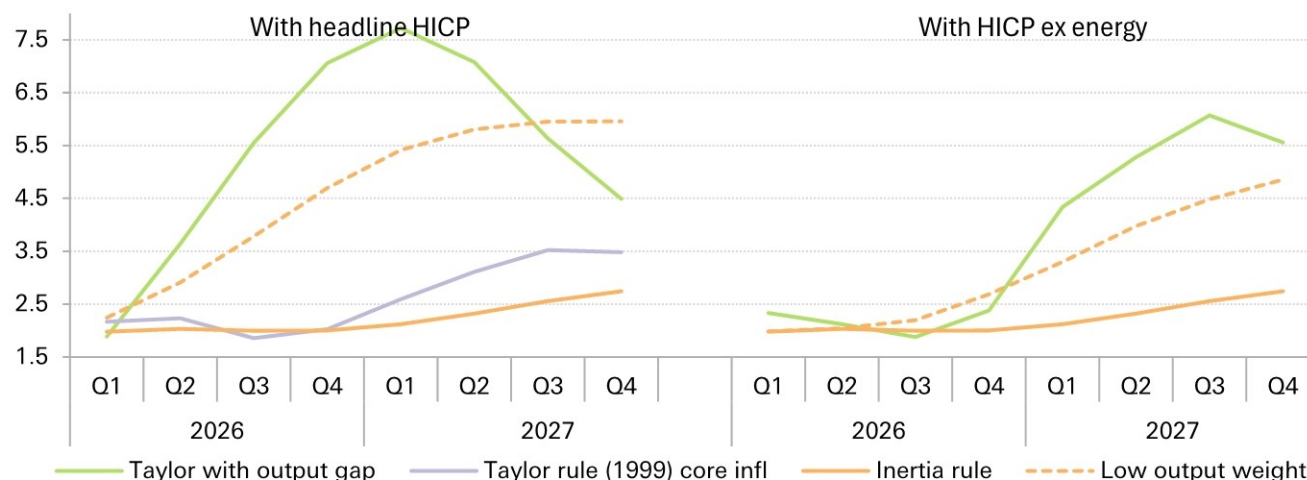
Source : Deutsche Bank Research, ECB, Haver Analytics, Bloomberg Finance LP.

Notes: Only a subset of policy rules can be analysed as the ECB does not publish unemployment data for their alternative scenarios. Colour of lines group rules based on the "type" of rule.





Figure 6: Severe scenario – policy rule prescriptions



Source : Deutsche Bank Research, ECB, Haver Analytics, Bloomberg Finance LP.

Notes: Only a subset of policy rules can be analysed as the ECB does not publish unemployment data for their alternative scenarios. Colour of lines group rules based on the "type" of rule.

**Caveat on policy rules under alternative scenarios:** It is important to keep in mind that the paths of inflation and output in the ECB's alternative scenarios are predicated on the same path of monetary policy (3 hikes) and fiscal policy as in the baseline. As the rules show, monetary policy should tighten under the adverse and severe scenarios, and be a little looser under the milder scenario (all relative to the baseline). Therefore, if any of these scenarios were to actually materialise, it is highly likely that monetary policy (and possibly fiscal policy) would respond differently to what the projections for inflation and GDP assume, which in turn would influence the actual outcomes of inflation and GDP. This means that we should keep in mind the policy rules give prescriptions for the specific path of inflation and GDP in the scenario, not how monetary policy should respond to the specific energy shocks underlying the alternative scenarios.

## How the policy prescriptions have evolved: March vs June staff projections

At the June meeting, the ECB staff revised up their forecasts for inflation noticeably in 2026-27, and revised down their growth only marginally over the projection.<sup>2</sup> [Figure 7](#) and [Figure 8](#) plot the difference in policy prescriptions based on the staff's June baseline projections and those based on the March baseline projections. This difference shows how the policy rules have evolved with the changes in inflation, unemployment, and growth outlooks under the two baseline projections.

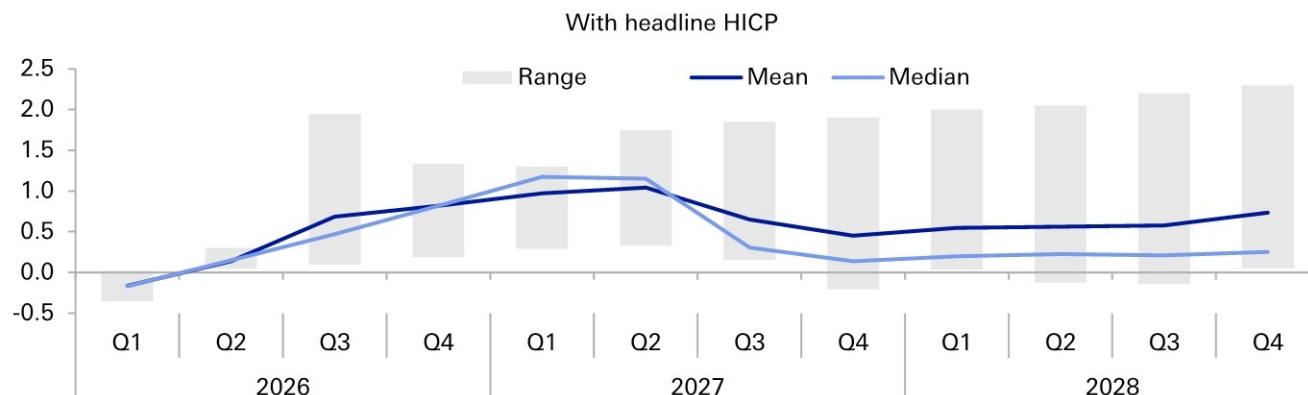
Because of the large number of rules, we present the range of differences (represented by the grey bars) alongside the average and median differences across all the rules in the suite.

<sup>2</sup> The underlying assumptions also changed, pricing in three policy hikes rather than two.



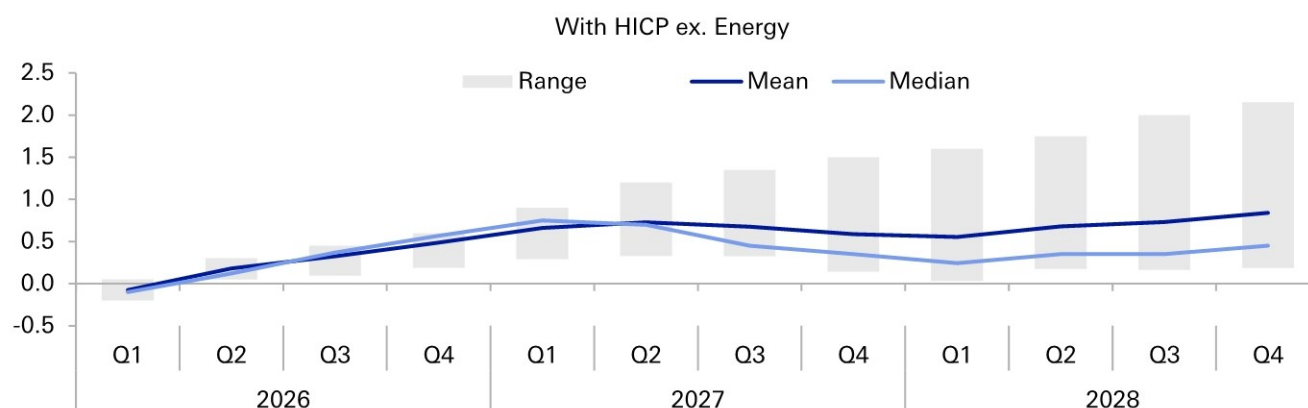


Figure 7: Difference between the policy rule prescriptions (pp) using the June staff projections vs the March projections – when headline HICP is used as the inflation measure in the rules



Source : Deutsche Bank Research, ECB, Haver Analytics, Bloomberg Finance LP. Notes: Range denotes the maximum-minimum difference in policy prescription between June and March staff projections

Figure 8: Difference between the policy rule prescriptions (pp) using the June staff projections vs the March projections – when HICP ex energy is used as the inflation measure in the rules



Source : Deutsche Bank Research, ECB, Haver Analytics, Bloomberg Finance LP. Notes: Range denotes the maximum-minimum difference in policy prescription between June and March staff projections

**Focusing on the results excluding energy prices, the revisions made to inflation and activity projections between the March and June staff projections would be consistent with around 50bp of additional tightening in early 2027, and 25-50bp further out in the projection (relative to the March policy rule prescriptions).** Between the March and June projections, market pricing has moved to price in one further hike over the next year, broadly in the range suggested by the rules.

## The ECB has not always followed these rules

Simple policy rules (like the ones in this report), serve as valuable benchmarks for assessing monetary policy but are not rigid prescriptions. While these rules provide a theoretical anchor based on core macroeconomic variables, actual policy rates often diverge. These divergences do not always necessarily reflect "policy mistakes". Central bankers incorporate a wider array of real-time data and forward-looking judgments not captured by stylized rules. But as benchmarks, the rules can



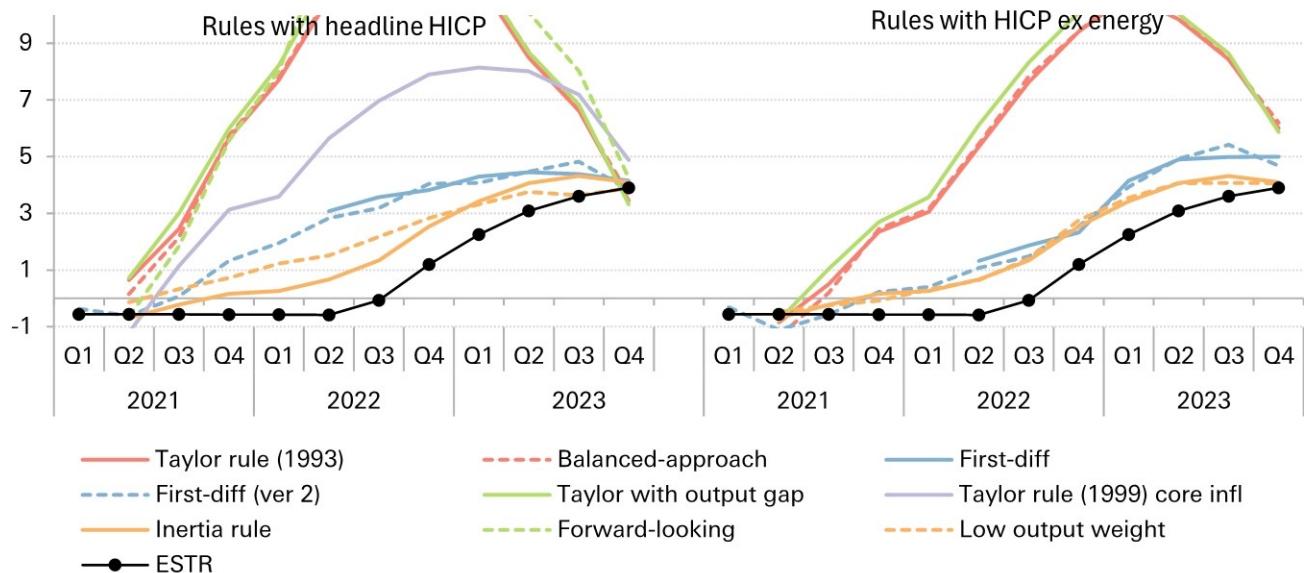
stimulate debate as to why actual policy deviates from the what the rules recommend as "good" policy actions.

Below we look at how actual interest rates compare to the policy prescriptions from the rules over the last few years. **The results show that the ECB does not follow any one specific rule.**

### 2021-2022 pandemic and energy shock

[Figure 9](#) plots the interest rate prescriptions from the policy rules over the period 2021-23 against the ESTR rate at the time. Even excluding the direct impact of energy prices, and assuming that interest rates are smoothed over time, the policy rules suggested interest rates should have started increasing by mid-2021, during the pandemic. The rates should then have further accelerated in early 2022 with Russia's invasion of Ukraine.

Figure 9: ESTR vs policy rule prescriptions (%) during the 2021-22 pandemic and energy shock



Source : Deutsche Bank Research, ECB, Haver Analytics, Bloomberg Finance LP. Notes: Colour of lines group rules based on the "type" of rule

Even though ESTR did not begin rising until mid-2022, the ECB argues it did not respond too late to the shock. Before it started hiking rates, the ECB tightened policy through the balance sheet – something the simple rules do not capture.<sup>3</sup>

One positive take away from [Figure 9](#) is that the first difference rules and policy rules with smoothing did a reasonable job at predicting the peak of the tightening cycle ahead of the actual peak.

### 2024 rate cutting cycle

[Figure 10](#) shows the policy rule prescriptions during the ECB's more recent cutting cycle (which started June 2024). The path of ESTR was broadly in the middle of the range of rules based on headline HICP throughout 2024. It then moved towards the

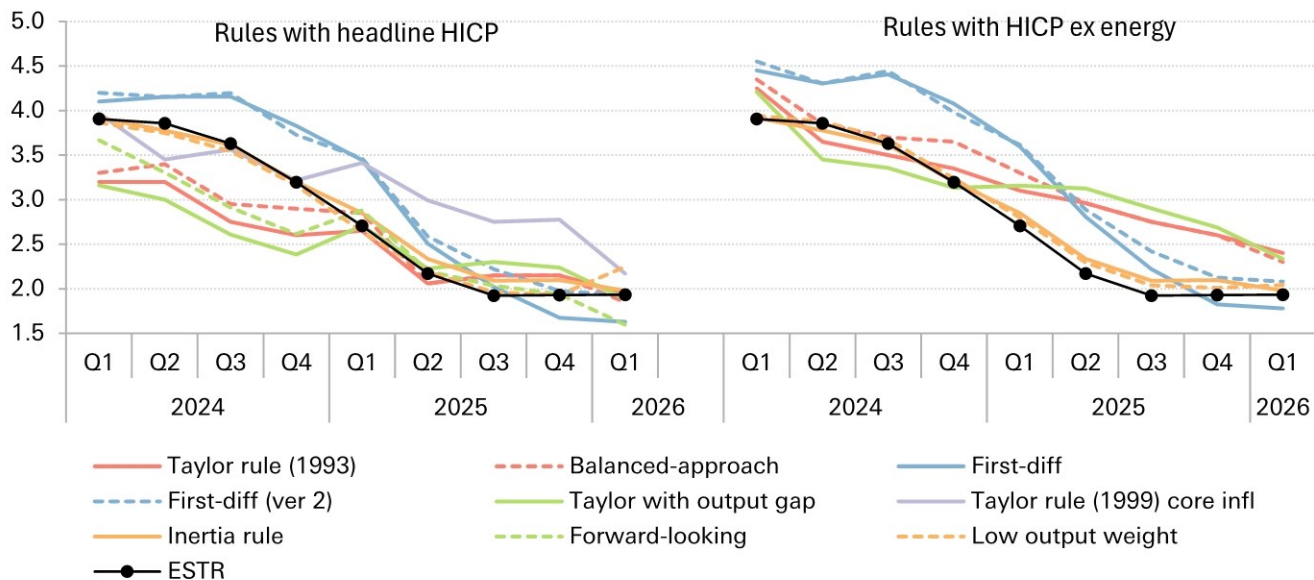
<sup>3</sup> In December 2021 the ECB decided to discontinue PEPP and taper APP throughout 2022.



lower end of that range in 2025. In terms of rules based on HICP excluding energy, ESTR remained close to the lower end of the range throughout the period.

**Overall the rules suggest the ECB's rate cutting cycle was justified by the inflation and activity data at the time.** But some of the ECB hawks might argue that the rate cutting cycle could have ended earlier than it did give the range of prescriptions in the second half of 2025 was skewed to the upside.

Figure 10: ESTR vs policy rule prescriptions (%) during the latest rate cutting cycle



Source : Deutsche Bank Research, ECB, Haver Analytics, Bloomberg Finance LP. Notes: Colour of lines group rules based on the "type" of rule





# Appendix 1

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